

Sarim Zafar

1001 Bay Street, Toronto, ON · zafarhsarim@gmail.com · 416-580-3152
sarimzafar.io · [🐙/sarimzafar](https://github.com/sarimzafar) · [in/sarimzafar](https://www.linkedin.com/in/sarimzafar)

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Applied Science: Honours Mechatronics Engineering, Cognitive Science Minor 2013 – 2018
Graduated with Distinction with a Cumulative Average of 84.42%

Relevant Courses

Graduate level Image Processing and Visual Communication, Autonomous Mobile Robotics, Autonomous Vehicles, Machine Intelligence, Pattern Recognition, Real-time Operating Systems, Digital Control Systems, Linguistics, Cognitive Processes

OBJECTIVE

My objective is to pursue graduate research in machine learning systems, developing novel methods for efficient training and inference of large-scale models. Leveraging my experience in distributed systems, where I've built real-time, fault-tolerant applications serving over 100 million users, I aim to address the cost efficiency challenges of large language and vision models, including high compute and data demands. My goal is to design training techniques that maximize model performance and resource efficiency, enabling strong results with significantly reduced computational and data requirements.

INTERESTS

Computer Vision, Deep Learning, Visual Language Models, Model Training and Inference, Autonomous Robotics, Artificial Intelligence, Generative AI, Distributed Systems and Computing, Designing Large-scale applications

TECHNICAL SKILLS

Languages — Python, Java, Scala, C++, Ruby, Go, JavaScript

Frameworks — Spring, Rails, NodeJs, MySQL, GCP, AWS, Elastic Search

RESEARCH PUBLICATION

Sharma, D., Zafar, S., Tizhoosh, H., Babaie, M. (2018). Facial Recognition with Encoded Local Projections. *IEEE-Symposium Series on Computational Intelligence 2018*. ([Text](#))

Soleiman, H., Zafar, S.(2018). Moving object detection using simple Background subtraction. ([Text](#))

WORK EXPERIENCE

Coursera March 2025 – Present
Staff Software Engineer – Technical Lead Toronto, ON

- Designed and developed distributed systems infrastructure enabling real-time enrollment options and custom pricing across global markets, designing fault-tolerant microservices, event-driven data pipelines to serve 100M+ learners with zero-downtime.
- Served as product and technical lead across pricing, subscriptions, and financial aid, aligning enterprise, consumer, architecture review, compliance, and support stakeholders to deliver complex, multi-quarter roadmaps on schedule despite reorgs and scope changes.
- Pioneered uncharted territory by building Coursera's regional pricing and localization capabilities from scratch, establishing new technical patterns for currency handling, auditing, and feature flagging that became organizational standards for international expansion.

- Led a team of 20+ engineers across product, platform, and payments to launch Coursera's freemium business model, navigating ambiguous requirements under tight deadlines. Built sophisticated freemium access controls balancing learner experience with revenue protection via policy engines, abuse detection, and conversion funnel instrumentation.
- Established a high-performance engineering culture through structured mentorship programs, comprehensive code review processes, and robust on-call practices, elevating code quality standards and reducing incident response times across multiple engineering teams.

Coursera

Senior Software Engineer

March 2022 – Feb 2025

Toronto, ON

- Architected and launched Coursera's revamped pricing engine, developing experiment-first pricing along with caching strategies that enabled rapid experimentation, generated \$1-2M annual incremental revenue, and reduced manual pricing updates from days to minutes through innovative pricing model redesign.
- Led large-scale platform migrations from legacy Scala systems to modern Java architectures across critical financial infrastructure, implementing dual-read systems, backfill processes, and data validation frameworks to ensure zero-downtime transitions while maintaining data integrity.
- Designed and implemented Financial Aid Ceiling system by building policy engines, approval workflows, and automated expiration mechanisms that streamlined aid distribution processes and projected \$4M additional annual revenue through improved operational efficiency.
- Built payment-processor-agnostic subscription platforms that integrate diverse billing systems (Google Play, Stripe, regional processors), creating reusable architectures that enabled international market expansion and flexible subscription management across multiple product tiers.
- Rapidly learned unfamiliar technical domains including billing system integration, payment processing workflows, and subscription lifecycle management, consistently delivering robust solutions under tight deadlines while establishing mentorship programs and comprehensive technical documentation.

Coursera

Software Engineer II

Sept 2020 – Feb 2022

Toronto, ON

- Led India Non-Recurring Payments (prepaid passes) end-to-end—from implementation and release to multiple experiment iterations—reviving a legacy product into Specialization Prepaid and leveraging ScheduledQueue for reminder workflows to increase LTV.
- Evangelized and technically led Stable Local Pricing, delivering consistent local-currency pricing by updating create-price endpoints, pricing engine logic, ORM fetches, and discount handling; authored technical docs, rollout plans, and extensive unit tests.
- Drove the Payments Database Migration from RDS to Aurora with data-services, becoming an early adopter of load testing and establishing a repeatable blueprint for future database upgrades and online migrations via gh-ost.
- Achieved the first month with 100% availability goals across all team services by leading reliability efforts, codifying best practices, and addressing cyclic latency and third-party outage playbooks (e.g., Razorpay incidents).
- Authored core engineering standards and mentorship artifacts (Consumer Java Style Guide, code-smell guidance for availability, database table protocols), while mentoring interns and driving design/code reviews across enrollment, pricing, and payments domains.

Ritual Technologies

Software Engineer II

Jan 2020 – Aug 2020

Toronto, ON

- Built menu ingestion platform to import menus from any POS into Ritual's catalog, using graph-based diffing to detect structural/content changes and apply targeted updates for accuracy and minimal downtime.
- Led *Similar to Merchant* search on Ritual, improving retrieval relevance and raising order conversion by 3% through query expansion, ranking tweaks, and iterative A/B testing.
- Drove *Sign in with Apple* integration to enable private, low-friction onboarding, increasing signup conversion while reducing credential management and fraud surface via platform authentication and relay email.
- Assumed project management responsibilities across releases, running design reviews, sprint planning, JIRA workflows, and test plan sign-off to de-risk launches and keep delivery on schedule.

Ritual Technologies

Software Engineer I

Dec 2019 – Dec 2020

Toronto, ON

- Overhauled the payments stack to meet SCA requirements and unblock the Europe launch, implementing 3DS2 flows, exemption handling, and end-to-end auth challenges to reduce declines and fraud exposure.
- Built a Java/Spring operations console to monitor inbound orders and surface remediation workflows, cutting time-to-resolution for support tickets and improving reliability during peak traffic.
- Refactored legacy modules and introduced rigorous unit and integration test coverage, stabilizing critical code paths and reducing regression risk across releases.

University of Waterloo @ Prof Hamid Tizhoosh

Jan 2018 — June 2018

Research Project Student

Waterloo, ON

- Conducted undergraduate research in facial recognition under Prof. Hamid Tizhoosh (Kimia Lab, University of Waterloo), focusing on handcrafted descriptors and robust feature extraction for unconstrained faces.
- Implemented end-to-end evaluation on benchmark face datasets, comparing ELP against LBP using chi-squared retrieval and SVM classification; demonstrated competitive accuracy with improved discrimination in comparable sub-image configurations.
- Authored experimental code, documentation, and analysis, ensuring reproducible results to validate the ELP approach against LBP and published the work in the IEEE-Symposium Series on Computational Intelligence 2018 ([Text](#))

University of Waterloo

Jan 2015 — April 2018

Teaching Assistant (6 terms)

Waterloo, ON

- Teaching Assistant for first year calculus, linear algebra, programming and circuits courses; at UW-WEEF lab.
- Conducted weekly tutorials, assisted with assignments/labs and provide feedback to students

Uber Technologies

May 2017 – Aug 2017

Software Engineer

San Francisco, USA

- Developed a Go/RPC based microservice to automatically name arbitrary [geographic clusters](#) around the world.

PROJECT

Automated Foosball playing robot ([Foosfighter](#))

Sept 2017 – April 2018

Python, C++, OpenCV

University of Waterloo

- Built an automated, opponent-grade foosball table capable of training human players, integrating real-time vision, control, and actuation to simulate competitive play and provide repeatable practice scenarios.
- Developed the vision pipeline in OpenCV to track and predict ball motion at real-time speeds, combining color-based segmentation with motion detection and background subtraction to maintain robust lock under occlusions and lighting changes.
- Implemented a Kalman filter for trajectory prediction, estimating position and velocity to anticipate ball location between frames and during brief visual dropouts, improving interception timing and control responsiveness.
- Wrote a custom thread-safe queue to decouple frame capture from processing, enabling concurrent I/O and inference; reduced initial pipeline latency by approximately 70% and stabilized frame throughput under load.
- [Demo video of Foosfighter in action.](#)

HONORS AND AWARDS

Best Technical Content, Mechatronics Design Symposium

UWaterloo

Team awarded for capstone project Foosfighter robot (\$1000).

2018

2x Engineering International Student Scholarship

UWaterloo

Awarded scholarship for high first year average (\$5000).

2014-15

University of Waterloo President's Scholarship

UWaterloo

Entrance award for high admission average (\$2000).

2013